

SHAILESH WASTI

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Education

The Pennsylvania State University

PhD Candidate in Energy Systems Engineering, Minor in Operations Research

Dissertation: Electricity Market Design with Storage under Uncertainty (Advisor: Prof. Mort Webster)

State College, PA

2027 (Expected)

University of Tennessee at Chattanooga

Master of Science in Engineering: Electrical (Power & Energy)

Thesis: Distributed Online Algorithms for Energy Management in Smart Grids. (Advisor: Prof. Vahid Disfani)

Chattanooga, TN

2020

Tribhuvan University, Kathmandu Engineering College

Bachelor of Engineering in Electrical Engineering

Kathmandu, Nepal

2015

Professional Experience

System Planning - PhD Intern

06, 2024 – 08, 2024

PJM Interconnection

Audubon, PA

- Developed long-term electricity demand and load growth scenarios to support power infrastructure planning and system reliability assessments
- Supported planning engineers with scenario-based modeling for infrastructure investment decisions

Lecturer

06, 2015 – 07, 2018

Kathmandu Engineering College

Kathmandu, Nepal

- Taught undergraduate courses in electrical machines, control systems, power utilization, and engineering materials
- Mentored student projects and edited newsletter of the department

Research Experience

Graduate Research Assistant

08, 2021 – Present

The Pennsylvania State University

State College, PA

- Developed large-scale optimization models of electricity markets for PJM Interconnection and ISO New England
- Designed coordination mechanisms for energy storage participation across day-ahead and real-time markets under forecast uncertainty to inform market design and regulatory decisions
- Developing pricing algorithms to enhance short-term operational efficiency and long-term investment efficiency for low-carbon power systems.

Graduate Research Assistant

08, 2018 – 04, 2021

University of Tennessee at Chattanooga

Chattanooga, TN

- Designed and modeled urban electric vehicle charging markets for Chattanooga
- Developed frameworks for the optimal allocation of virtual inertia for high-DER power systems

Publications and Presentations

- **S. Wasti**, A. Retna Kumar, S. Varghese, A. Giacomoni, and M. D. Webster, “Energy Storage State-of-Charge Management in Real-Time Markets,” IEEE PES General Meeting, July 2025.
- **S. Wasti**, A. Retna Kumar, S. Varghese, A. Giacomoni, and M. D. Webster, “Energy Storage Management in Electricity Markets under Intra-Day Uncertainty,” under review, IEEE Trans. Energy Markets, Policy and Regulation.
- **S. Wasti (Invited Speaker)**, A. Giacomoni, A. Retna Kumar, M.D. Webster, “Energy Storage Management in Electricity Markets under Intra-Day Uncertainty,” EPRI ISO/RTO Emerging Technologies Market Modeling Working Group Meeting, Oct. 2025.
- **S. Wasti**, P. Macedo, S. Afshar, J. Griffin, V. R. Disfani, and P. Siano, “Distributed Dynamic Algorithm for Energy Management in Smart Grids,” in *Decentralized Frameworks for Future Power Systems*, Elsevier, 2022.
- F. Mohamed, **S. Wasti**, S. Afshar, P. Macedo, and V. Disfani, “MMC-Based Distributed Maximum Power Point Tracking for Photovoltaic Systems,” IEEE PES General Meeting, 2020.

Technical Skills

Optimization & Modeling: GAMS, JuMP, Pyomo, CVX, CVXPY, MATLAB Simulink, AnyLogic

Programming & Data Analysis: MATLAB, Julia, R, Python, Tableau, Excel, C/C++

Developer Tools: Git/GitHub, LaTeX, HPC clusters (Slurm, PBS)

Graduate Coursework: Convex Optimization, Stochastic Optimization, Integer Programming, Economic Analysis of Energy Markets, Power System Operations, Applied Data Mining and Statistical Learning, Microeconomic Analysis